Advanced GANs

Study how architectural refinements improve high-resolution adversarial generation, controllability, and latent representations, rather than treating every GAN extension as a new unrelated method.

Original study notes based on David Foster, Second edition, 2023. Book pp. 267-296. Source APIs, services, experiments, and model comparisons reflect the book's publication period. These are explanatory summaries, not verbatim reproductions or a claim that every source detail is retained.

Progressive resolution → Style-controlled synthesis → Improved normalization → Attention and scale → Discrete visual tokens

Detailed Notes

ProGAN and progressive training

Source: book pp. 267-276 / PDF p. 295

ProGAN begins training at low resolution, then adds higher-resolution generator and discriminator layers. Fade-in blending introduces new paths gradually so existing coarse structure is not abruptly disrupted.

Early layers learn large-scale structure; later stages add detail. Equalized learning-rate ideas, normalization, minibatch diversity signals, and stable transition schedules support training. Inspect outputs across resolutions and remember that sharp local texture does not establish correct global geometry. Progressive growing is a historical solution, not an indispensable ingredient of every successful image GAN.

StyleGAN mapping and synthesis networks

Source: book pp. 277-280 / PDF p. 305

StyleGAN maps input z through an MLP into an intermediate latent w, which controls synthesis at multiple resolutions. This intermediate representation can be easier to manipulate than the original prior space.

Adaptive instance normalization modulates feature statistics with style-dependent scale/shift, while separate noise inputs add stochastic fine detail. Style mixing uses different styles at different layers to discourage entanglement. Coarse and fine controls often influence different visual scales, but semantic separation is empirical rather than guaranteed. Output analysis should include interpolation, repeated styles with changed noise, and edits.

StyleGAN2 modulation and demodulation

Source: book pp. 281–282 / PDF p. 309

StyleGAN2 addresses artifacts associated with the earlier normalization design. Style-dependent weight modulation scales convolutional weights; demodulation normalizes their expected output scale instead of directly normalizing each resulting feature map as before.

This reduces characteristic artifacts while retaining style control. Track which dimensions of the convolutional weight tensor receive modulation and which axes determine normalization. It is not equivalent to adding arbitrary batch normalization after every convolution. Implementation details and regularization interact with stability.

Path-length regularization and no progressive growing

Source: book pp. 283–286 / PDF p. 311

Path-length regularization encourages more uniform sensitivity of image changes to steps in the intermediate latent space. It improves the conditioning of the latent-to-image mapping and editing behavior, but is not a general guarantee of causal disentanglement.

StyleGAN2 replaces the original progressive-growing training strategy with a redesigned architecture and multiscale skip/residual paths. Compare output artifacts and controllability rather than assuming a higher model version solves every data or training defect. Separate generator-side regularization from discriminator penalties.

Self-attention GAN and BigGAN

Source: book pp. 286–288 / PDF p. 314

SAGAN adds self-attention so distant image regions can influence one another, helping coordinate global relationships that local convolutions model less directly. Normalization and stable critic behavior remain important.

BigGAN scales class-conditional generation with architectural and optimization refinements, large batches, and class information. Truncating the latent sampling distribution can improve apparent fidelity at the expense of diversity. Always evaluate the fidelity/coverage trade-off instead of reporting only hand-picked sharp images.

VQ-GAN: discrete latent image tokens

Source: book pp. 289-291 / PDF p. 317

A vector-quantized encoder maps images into a grid of discrete codebook entries. A decoder reconstructs images, while perceptual/adversarial losses encourage useful visual detail. A commitment/codebook mechanism aligns continuous encoder outputs with selected entries.

A separate autoregressive model can learn distributions over these compact visual-token grids, reducing sequence length relative to pixels. Distinguish training the image tokenizer from training the prior over its codes. Poor codebook usage, collapse, or reconstruction distortion limits the downstream generator.

ViT VQ-GAN and synthesis of ideas

Source: book pp. 292-296 / PDF p. 320

ViT VQ-GAN introduces transformer-based components and codebook/training refinements for visual quantization. It illustrates combining discrete representation learning with expressive token models rather than choosing permanently between GANs and transformers.

Compare reconstruction quality, code utilization, generation quality, compute, and controllability. Improvements arise from specific architectural/optimization choices under particular datasets. Reproducing a named model requires the matching recipe, not just importing a layer with a similar name.

Revision Check

1. How does progressive fade-in work?
2. What distinguishes z and w in StyleGAN?
3. Why separate stochastic noise from style?
4. What does demodulation normalize?
5. How can truncation improve fidelity while hiding lost diversity?
6. Which two learning problems occur in VQ-GAN plus an autoregressive prior?

High-quality image generation combines stable training, global coordination, and usable representations. Analyze the mechanism behind each named extension.

Source links refer to the supplied PDF. Every contents-page topic is mapped in coverage.md. Consult the book for complete implementations, figures, exercises, citations, and experiment evidence.